

The Platonic Representation Hypothesis

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The Platonic Representation Hypothesis

Observation

“Neural networks, trained with different objectives on different data and modalities, are converging to a shared model in their representation spaces.” Huh et al. (2024)

Proposed Explanation

There must be some underlying shared reality responsible for this convergence.

Additional claim

The model NNs are converging on is also (to some extent) shared by the human mind.

Why is there convergence?

- Contravariance Principle: The more tasks we must solve, the fewer functions satisfy them all (Cao and Yamins 2021).
Example: Autoregressive language modeling can be seen as optimizing a diverse set of tasks (Radford et al. 2019).
- Simplicity Bias: Deeper/bigger models have a stronger simplicity bias (= smaller solution space)
- Capacity Hypothesis: There is an optimal solution and models with sufficient capacity will find it: Hu et al: “the representation we are converging toward is a statistical model of the underlying reality that generates our observations.”

What is a “statistical model of underlying reality”?

“In a particular idealized world, we show that a certain family of learners will converge to a representation whose [similarity measure] is equal to the pointwise mutual information (PMI) function over the underlying events (Z) that cause our observations, regardless of modality.”

$$\text{sim}(f_{\text{red}}, f_{\text{orange}}) = \text{sim}(\text{●} \text{ ●}) = \text{PMI}(z_{\text{red}}, z_{\text{orange}}) + \text{const}$$

Pointwise Mutual Information (PMI)

Contrastive learner: Train models by maximizing agreement between different views of the same data instance (positives) while pushing away representations of different instances (negatives).
Conjecture: More like human perceptual learning.

Assume a contrastive NN learner that models what observations cooccur. Positive pair: two observations occur close in time.
Negative pair: observations sampled from anywhere in time.

$$\text{PMI}(x, y) = \log \frac{P(x, y)}{P(x) P(y)}$$

A key claim of the formal proof is the loss functions of many contrastive learners optimize PMI. (Note: Bert is an example of a **predictive** learner).

Ding-an-sich?

The Platonic Representation Hypothesis

Neural networks, trained with different objectives on different data and modalities, are converging to a shared statistical model of reality in their representation spaces.

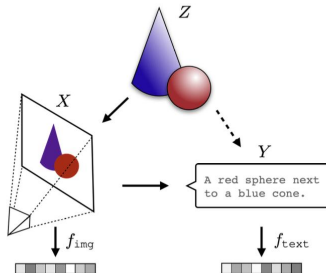


Figure 1. The Platonic Representation Hypothesis: Images (X) and text (Y) are projections of a common underlying reality (Z). We conjecture that representation learning algorithms will converge on a shared representation of Z , and scaling model size, as well as data and task diversity, drives this convergence.

Three concepts for model alignment

Representation

a function $f : \mathcal{X} \mapsto \mathbb{R}^n$ that assigns a feature vector to each input in some data domain $\mathcal{X}$

Kernel

$$K : \mathcal{X} \times \mathcal{X} \mapsto \mathbb{R}$$

$$K(x_i, x_j) = \langle f(x_i), f(x_j) \rangle$$

$$K \in \mathbf{K}$$

Kernel Alignment Metric

$$m : \mathbf{K} \times \mathbf{K} \mapsto \mathbb{R}$$

measures the similarity between two kernels

Cross-modal models

For cross-modal domains, define $(x_i, y_i) \in \mathcal{X}$ as a sample from the data distribution $\mathcal{X}$, e.g. 1024 samples from WIT [Wikipedia-based Image Text](Srinivasan et al. 2021), $\dots x_i \sim y_i$.

Let $\{x_i, y_i\}_{i=1}^b$ be a mini-batch sampled from this data distribution and given two model representations f and g , let the corresponding feature vectors be: $\phi_i = f(x_i)$ and $\psi_i = g(y_i)$, where the collection of these features are denoted as $\Phi = \{\phi_1, \dots, \phi_b\}$ $\Psi = \{\psi_1, \dots, \psi_b\}$.

The KNN kernel (similarity measure)

Find the model representations of each text x and each image y ($f_{\text{text}}(x)$ and $f_{\text{img}}(y)$). Find the k -nearest neighbors ($k = 10$) of $f_{\text{text}}(x)$ and the k -nearest neighbors of $f_{\text{img}}(y)$:

$$\begin{aligned}K_{\text{img}}(i, j) &= \langle f_{\text{img}}(x_i), f_{\text{img}}(x_j) \rangle \\K_{\text{text}}(i, j) &= \langle f_{\text{text}}(y_i), f_{\text{text}}(y_j) \rangle\end{aligned}$$

This is inner-product (which appears to be dot product). If the vectors ($f(x_i)$) are normalized, this would be equivalent to cosine.

Kernel Alignment metric (model alignment)

For each vector pair (ϕ_i, ψ_i) , we compute the respective nearest neighbor sets $S(\phi_i)$ and $S(\psi_i)$:

$$\begin{aligned} \text{dknn}(\phi_i, \Phi \setminus \phi_i) &= S(\phi_i) \\ \text{dknn}(\psi_i, \Psi \setminus \psi_i) &= S(\psi_i), \end{aligned}$$

where dknn returns the set of indices of the k -nearest neighbors (according to the model kernel).

$$m_{NN}(\phi_i, \psi_i) = \frac{1}{k} |S(\phi_i) \cap S(\psi_i)|$$

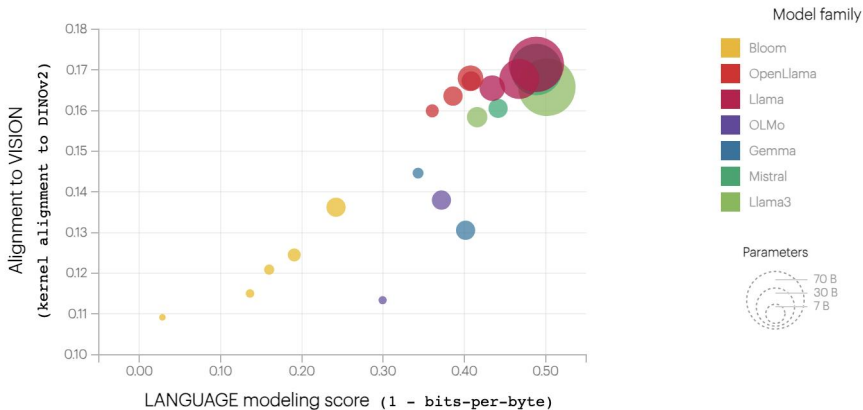
where $|\cdot|$ is the size of the intersection. Code [here](#)

The basic plausibility argument

An indicator: “[T]he rising number of systems built on top of pre-trained foundation models. These models are becoming standard backbones across a growing spectrum of tasks. Their versatility across numerous applications implies a level of universality in the way they represent data.”

But **Do different foundation models arrive at the same representation?**

Multimodal alignment: bigger models are more aligned (LLMs and DINOv2)



Summary of experiments/arguments

- Huh et al. cite a number of studies showing multi-modal synergy: Jointly training vision and language models improves both, especially Achiam et al. (2023). Citation count: 27,069 [Google Scholar]
- Denser captioning (more words) improves alignment.
- The better aligning language models perform better (Hellaswag: common sense; GSM8K: math; language modeling score)
- Models are increasingly aligning to brains; a lot of literature (Section 2.4)
- Abdou et al. (2021) discovered that color distances in learned language representations, when trained to predict cooccurrences in text, mirror human perceptions of these distances.
- Takeaway: Convergence of LLMs, Visual systems, human perceptions.

Incommensurables

What image does

*I believe in freedom of
speech*

converge to?

What sentence does

*the experience of a total
solar eclipse*

converge to?

Loopholes; Conclusions

- An observation is

$$\text{obs} : \mathbf{Z} \mapsto \cdot$$

where $\cdot$ is some measurement space (a mix of pixels, sounds, mas, torque, words?). Mathematical argument requires obs to be bijective and assumes discrete $\mathbf{Z}$.

- Language and Vision models align to an alignment score of .16 (with a theoretical max of 1). So that leaves open the possibility that no better score can be achieved, which leaves open the possibility of no single underlying model to converge on.
- Even a score of 1 would be achieved only with respect to samples from datasets that in turn make limited (and noisy) commitments to what should align: Caption X with picture Y (WIT).
- Language models learn things about the world (Bowman 2024).

Bibliography I

Abdou, Mostafa, Artur Kulmizev, Daniel Hershcovich, Stella Frank, Ellie Pavlick, and Anders Søgaard. 2021.

Can language models encode perceptual structure without grounding? a case study in color.

In Proceedings of the 25th conference on computational natural language learning, 109–132.

Achiam, Josh, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. . 2023.

Gpt-4 technical report.

arXiv preprint arXiv:2303.08774.

Bowman, Samuel R. 2024.

Eight things to know about large language models.

Critical AI 2(2).

Bibliography II

Cao, Rosa, and Daniel Yamins. 2021.

Explanatory models in neuroscience: Part 2 – constraint-based intelligibility.

Huh, Minyoung, Brian Cheung, Tongzhou Wang, and Phillip Isola. 2024.

Position: The platonic representation hypothesis.

In *Forty-first International Conference on Machine Learning*.

Radford, Alec, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. . 2019.

Language models are unsupervised multitask learners.

OpenAI blog 1(8):9.

Bibliography III

Srinivasan, Krishna, Karthik Raman, Jiecao Chen, Michael Bendersky, and Marc Najork. 2021.

Wit: Wikipedia-based image text dataset for multimodal multilingual machine learning.

In Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval, 2443–2449.